

Company: MSCI
Job Profile: Technology & Data Operations Intern
Offer Type: Internship (Performance based PPO)
Internship Stipend: Rs. 60,000/-
CTC: 14.68 LPA (Base 13.2 LPA)
Location: Mumbai / Pune
Process Date: OA : 11th July Interviews : 15th July

Placement Process:

Online Assessment(75 mins, 2 Coding Qs, 1 SQL Qs)

Interviews:

1. Technical Round 1
2. Technical Round 2
3. HR Round

1. **Online Assessment** [Around **240 candidates** appeared for this round]

There were 2 DSA questions and 1 SQL question in the online assessment and the time duration was 75 mins.

Q1 : Given an array of integers, find the sum of elements of all possible subarrays of the array.

For example, if the input array is [1, 2, 3], then the sum of all possible subarrays is 20.

The subarrays are: [1], [2], [3], [1, 2], [2, 3], [1, 2, 3]

Sum = 1 + 2 + 3 + 3 + 5 + 6 = 20

Here, I followed a brute-force approach by calculating all possible subarrays and maintaining the sum of all the subarrays.

I was able to pass **14 out of 15** test cases.

Q2 : Given an array of integers, the task was to reach the end of the array while maximizing the score based on allowed jumps.

There were two jump conditions:

1. I could take **one step to the immediate right**, i.e., $i \rightarrow i+1$
2. I could take a jump of length p such that p is a **prime number ending with 3** (e.g., 3, 13, 23, etc.).

The goal was to reach the end of the array while collecting the maximum possible score (i.e., sum of visited elements).

I followed a **dynamic programming (DP)** approach and used **memoization** to avoid recomputation. At each index, I considered both the one-step jump and all valid p jumps. I recursively computed the maximum score possible from that point onward, and memoised the results.

Q3: This was a SQL question involving two tables related to voter information.

The goal was to return a summary of each state, showing the names of voters who voted in that state along with the number of times each voter voted.

Given Tables:

Voters:

voter_id | name

-----|-----

1 | Tanvi

2 | Gaurang

3 | Drishya

4 | Rahul

Votes:

voter_id | state

-----|-----

1 | California

2 | California

2 | California

1 | California

4 | Texas

Expected Output :

state | voters_summary

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California | Tanvi x 2, Gaurang x 2

Texas | Rahul x 1

The question was supposed to be solved using GROUP_CONCAT()

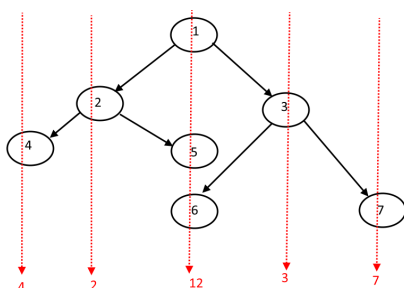
I couldn't fully solve the question during the interview due to time constraints. Try to solve the DSA questions fully first as they have the most weightage to get selected for interviews.

2. Technical Interview 1 [39 candidates were shortlisted for this round.]

The round lasted for approximately 50 minutes.

The interviewer began by introducing himself, and then asked me to give a brief introduction. After that, I was asked **two DSA questions**.

Q1: Vertical Sum of a Binary Tree : The question was based on **binary trees**. I was asked to compute the **sum of all elements in the tree that lie along the same vertical line**, and return an **array of sums for each vertical line**.



For example, the output was expected to look something like: [4, 2, 12, 3, 7].

I suggested using a hash map to store the sum of nodes at each horizontal distance (HD). I used $HD = 0$ for the root, $HD - 1$ for the left child, and $HD + 1$ for the right child. As we traverse the tree, we keep adding values to the same horizontal distance key in the hash map. I mentioned that since Python dictionaries are unordered, we'd need to sort the keys before generating the final output array and coded up the entire solution on paper. I was asked to perform a dry run and the interviewer was satisfied with my solution and moved on to the next question.

Q2 : The second question was a standard problem on detecting a cycle in a directed graph. I approached the problem using a Depth First Search (DFS) technique. To handle the detection, I used two boolean arrays: one to keep track of all the nodes that had been visited globally, and another to track the nodes in the current DFS recursion path. I explained this logic clearly and proceeded to code the solution during the interview. I was asked to perform a dry run and the interviewer was satisfied with my approach.

After the DSA questions, the interviewer shifted the focus to my projects and began going through my resume. He asked me to briefly explain the projects I had worked on and the technologies I used. Following that, he delved deeper into the topic of authentication, asking both conceptual and implementation-related questions. He wanted to understand how I had implemented authentication in my projects and how the flow works in a typical web application. Specifically, he asked about JWT (JSON Web Tokens) and its structure. The questions included:

Q1 : What is present in the **JWT header structure**?

Q2 : What is the complete **authentication flow**, **how** the token is generated, returned to the client, stored, sent in subsequent requests, and how token expiry (token window) is handled?

Q3 : How can we implement **role-based access control**? Here I answered that we can use wrappers or decorators to restrict access to api endpoints and maintain security and role based access. Then there were several questions on databases.

The interview concluded with the interviewer asking me if I had any questions for him. I took this opportunity to ask about the advancements in AI and how they are being leveraged in the finance domain, where both high volumes of data and data integrity are critically important. The interviewer answered the question briefly and the interview ended.

3. Technical Interview 2 [Around 20 candidates were shortlisted for this round.]

The second interview lasted for about an hour. The interviewer began by introducing himself and then asked me to introduce myself. I gave a brief overview of my background, skills, and projects. He then asked me which topics in DSA I was most comfortable with. I mentioned that I enjoy working with graphs and heaps. Following that, he asked me to explain real-life applications of graphs, which I answered with relevant examples like in maps for determining shortest paths.

He proceeded to ask a DSA question, which was a variation of Dijkstra's algorithm. I was asked to code the solution on paper. I wrote the code and explained the entire logic. During the explanation, he asked several detailed questions about each part of my code—why I used heaps, how I handled initialization, and the overall structure of the solution.

He then asked me to think through potential edge cases and confirm whether my code handled them. After that, he asked me to do a dry run of the code. I walked him through the step-by-step execution, and he was satisfied with my explanation and approach. After the DSA question, the interviewer briefly moved on to my resume. He asked about my participation in hackathons and what I had implemented in one of them. He then asked about the projects listed in my resume, including their purpose and technical implementation.

Since I had mentioned LangChain in my resume, he asked several questions related to it. These included: what is LangChain, why is it used, what is LangGraph, why not build directly without LangChain, and what advantages it offers. I answered these questions based on my project experience. Next, he asked me to explain another project I had worked on, including the technologies I used. He then shifted focus to OOP concepts and asked questions about the static keyword, implementation of singleton classes, encapsulation, and interfaces.

After that, he asked me two SQL questions. He initially provided two tables: student and subject, and asked a medium-level SQL query involving joins, which I solved successfully. Then he added a third table called department and asked a follow-up query. I was able to solve all three queries completely, and the interviewer seemed satisfied.

Then he asked a few questions related to React like to explain the component structure, then explain the two hooks usestate and useeffect. Then he asked about Virtual DOM and working of virtual DOM and how it improved efficiency. Then he asked some general topics like MongoDB and databases. He asked me to explain the differences between SQL and NoSQL databases, and in what scenarios I would choose one over the other. He also asked me to give two real-world examples where I would use both types, which I was able to answer.

Finally, he asked me about different types of NoSQL databases beyond key-value stores. I admitted I wasn't aware of all the types. He then asked about indexing in DBMS and whether indexing can be done in NoSQL databases as well.

The interview concluded after this discussion and I asked about how AI is leveraged in finance. He answered briefly and the interview concluded.

4. **HR Round** [8 candidates were shortlisted for this round.]

The HR round was very relaxed and conversational. The interviewer was warm and made me feel comfortable from the beginning. It felt more like a discussion than a formal interview and lasted around 15 minutes.

1. Introduce yourself
2. Mention something interesting that is not in your resume
3. What do you like to do besides tech?
4. Explain two of your projects
5. Describe a time you worked in a team
6. Do you prefer working individually or in a team?
7. Where do you see yourself in 5 years?
8. Are you planning for higher studies?

For the question about higher studies, I responded that I'm not planning to pursue them. I shared that I believe gaining hands-on industry experience is the right path for me.

At the end of the interview, she asked if I had any questions for her. I asked about MSCI's commitment to ESG and climate initiatives, and how those values are integrated into the company's work culture. She appreciated the question and gave a thoughtful response before concluding the interview.

[5 students were offered the job]

Useful Tips:

1. DSA Preparation

Start DSA preparation early on. Try to consistently practice at least 1–2 questions every day. I personally followed Striver's SDE Sheet, which is very well-structured and covers most interview-relevant topics.

<https://takeuforward.org/strivers-a2z-dsa-course/strivers-a2z-dsa-course-sheet-2>

Always try to solve questions yourself first. If you're stuck, watch the video explanation but make sure to code it yourself afterwards. Keep revising important questions and try to identify and remember common problem patterns across topics.

2. Core Subjects – Java and OOPs

I started with the basics of Java. Here's a link I referred to:

https://youtu.be/eIrMbAQSU34?si=_Vq7MalRpZaXJsO9

For OOPs concepts, I followed Kunal Kushwaha's playlist, which explains everything in a beginner-friendly manner. Additionally, I read articles on GeeksforGeeks, which helped reinforce key concepts like encapsulation, inheritance, polymorphism, and abstraction.

<https://youtu.be/BSVKUk58K6U?si=NQVHOYRXLAGVkir6>

<https://www.geeksforgeeks.org/java/object-oriented-programming-oops-concept-in-java/>

3. DBMS (RDBMS)

For RDBMS, I watched CodeHelp by Love Babbar's one-shot video. It covers almost all the concepts required for interviews.

https://youtu.be/dl00fOOYLOM?si=IE8ak_4YztXdYTOF

4. SQL

I learned the basics from W3Schools and practiced extensively on LeetCode. I completed all 50 questions from the SQL 50 list and also solved 10–15 additional medium/hard-level questions to get more comfortable

<https://www.w3schools.com/sql/default.asp>

<https://youtu.be/OT1RErkfLNQ?si=1Ctjw2Ud6hQRm-X->

<https://leetcode.com/studyplan/top-sql-50/>

5. Operating Systems & CN

For OS and Computer Networks, basic articles from GeeksforGeeks are more than enough. Focus on understanding concepts like process/thread, scheduling, memory management, TCP/IP, and HTTP. I wasn't asked this in interviews but some of my friends were asked about these subjects.

6. System Design in OOPs

In some interviews, you're asked to design systems like a Library Management System, Parking Lot, or ATM Machine using OOPs concepts. So, be prepared to design small-scale systems applying classes, objects, inheritance, and design principles. Practice dry-running a few systems beforehand.

7. Final Interview Tips

Before every interview, research the company, their culture, products, and latest updates. This not only helps you frame better answers but also helps you ask relevant questions at the end of the interview, which leaves a strong impression. Prepare an introduction which basically summarizes your resume. Know your projects and your resume very thoroughly.

If you have any questions, I'll be more than happy to help. You can reach out to me directly on WhatsApp or LinkedIn.

All the best